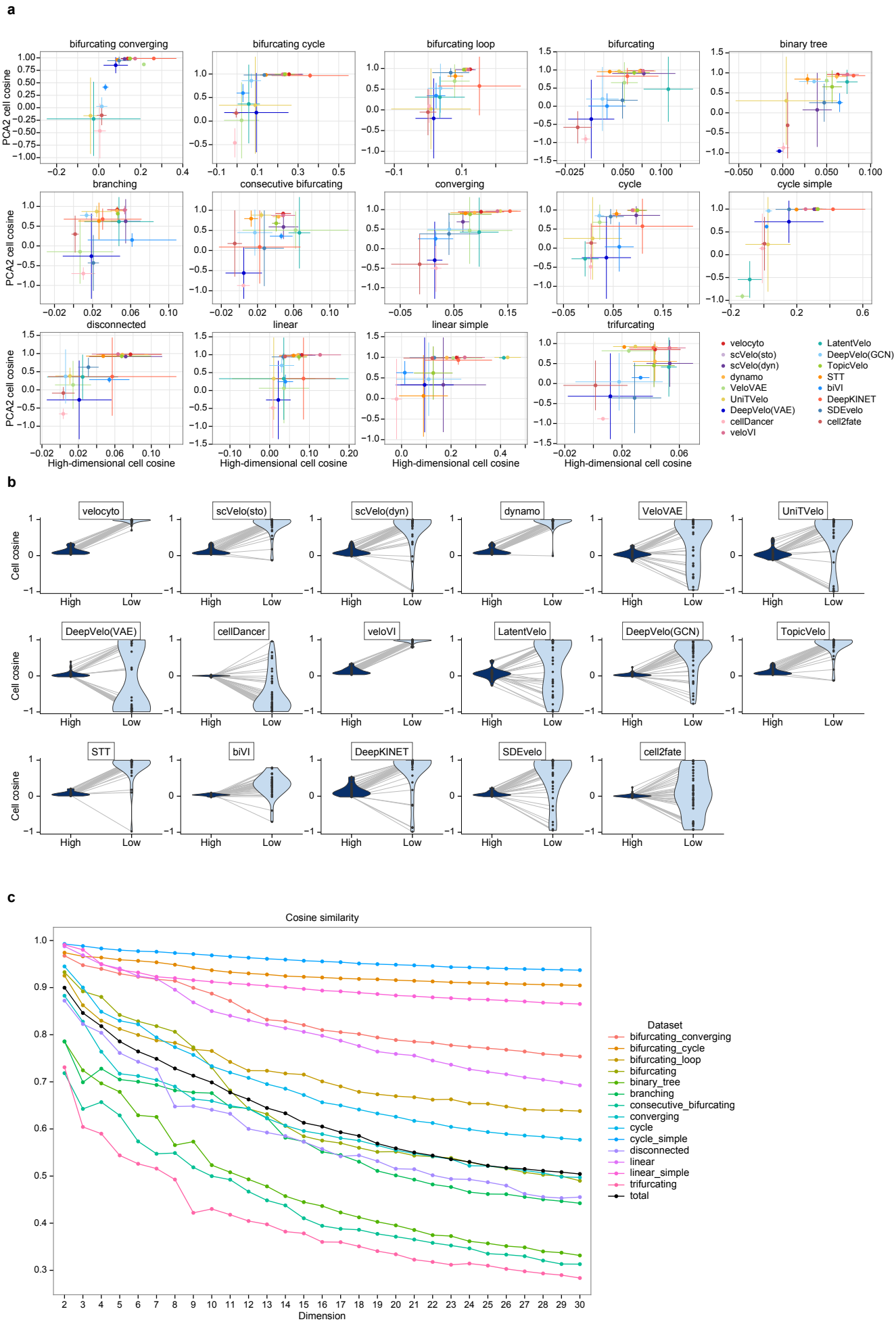

a
bifurcating converging
bifurcating cycle
bifurcating loop
bifurcating
binary tree
branching
consecutive bifurcating
converging
cycle
cycle simple
disconnected
linear
linear simple
trifurcating
PCA2 cell cosine
High-dimensional cell cosine
velocyto
scVelo(sto)
scVelo(dyn)
dynamo
VeloVAE
UniTVelo
DeepVelo(VAE)
cellDancer
veloVI
LatentVelo
DeepVelo(GCN)
TopicVelo
STT
biVI
DeepKINET
SDEvelo
cell2fate
b
Cell cosine
High
Low
c
Cosine similarity
Dimension
Dataset
bifurcating_converging
bifurcating_cycle
bifurcating_loop
bifurcating
binary_tree
branching
consecutive_bifurcating
converging
cycle
cycle_simple
disconnected
linear
linear_simple
trifurcating
total

Extended Data Fig. 5 | High- and low-dimensional velocity evaluation. a, Scatter plots separated by the 14 dyngen topologies. Each point summarizes one method across available matched seed datasets (mean and sample SD); high-dimensional and PCA2 scores are cell-level cosine similarities. **b,** Paired high/low-dimensional violin plots, faceted by method; lines join the same dataset, with dataset medians shown as points. **c,** Cosine similarity between predicted and simulated true RNA velocity in PCA spaces of 2–30 dimensions. Each colored point is the median of finite method–dataset cell-level cosine scores within one topology at that dimension; the black total curve is the median across all 42 datasets and available methods. These are projected velocity-accuracy scores, distinct from the high/low-dimensional score-vector consistency in Fig. 3d. Seventeen available configurations are included; missing scores are excluded without imputation.